

DAY 4- GPU AND COOLING SYSTEMS:

Tasks

1.

Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

The GPU (Graphics Processing Unit) is responsible for processing and displaying graphics, videos, games and other visual content. It can either be integrated into the processor or be a separate dedicated graphics card.

Method 1: Using Windows Task Manager

1. Start your laptop/desktop.
2. Press Ctrl + Shift + Esc to open Task Manager.
3. Click the Performance tab.
4. Look for GPU 0, GPU 1, etc.
5. Click on the GPU.
6. Note down the GPU name/model shown in the top-right corner.

Method 2: Using Device Manager

1. Press Windows + X.
2. Select Device Manager.
3. Expand Display adapters.

4. The GPU model will be displayed there.

For example:

NVIDIA GeForce RTX 4060 Laptop GPU

Determine whether it is integrated or dedicated

Type	Meaning	Example
Integrated GPU	Built into the CPU/processor and generally shares system RAM	Intel UHD Graphics, AMD Radeon Graphics
Dedicated GPU	Separate graphics processor/card with its own video memory (VRAM)	NVIDIA GeForce RTX 4060, AMD Radeon RX 7600

Assignment Answer Format

Fill this in using the information from your computer:

Detail	Answer
GPU Brand	_____
GPU Model Number	_____
GPU Type	Integrated / Dedicated
Where identified	Task Manager / Device Manager

Example

GPU Brand: NVIDIA

GPU Model: GeForce RTX 4060

GPU Type: Dedicated

2.

Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.

The cooling system prevents computer components such as the CPU and GPU from overheating. It usually uses fans and heatsinks, while some high-performance systems use liquid cooling.

Step 1: Open the PC or use a simulator

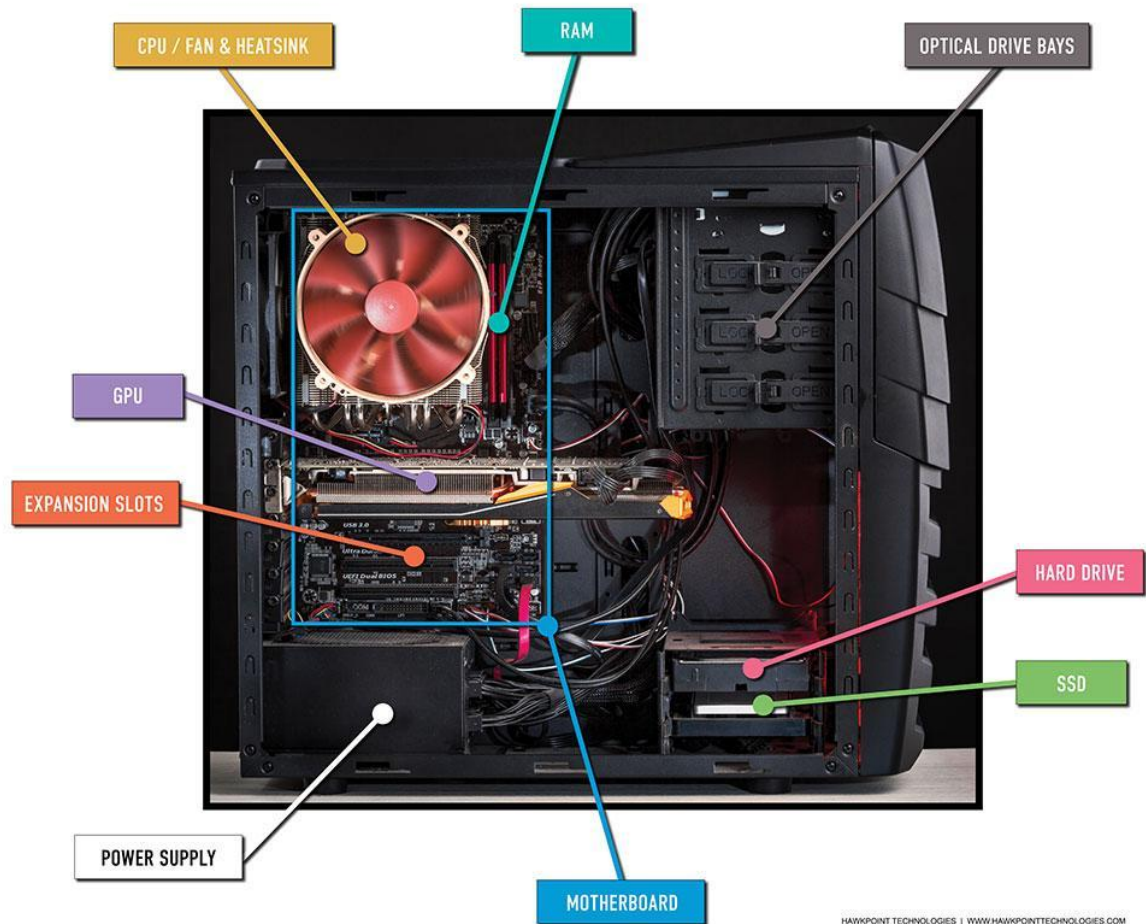
If you're using your own desktop:

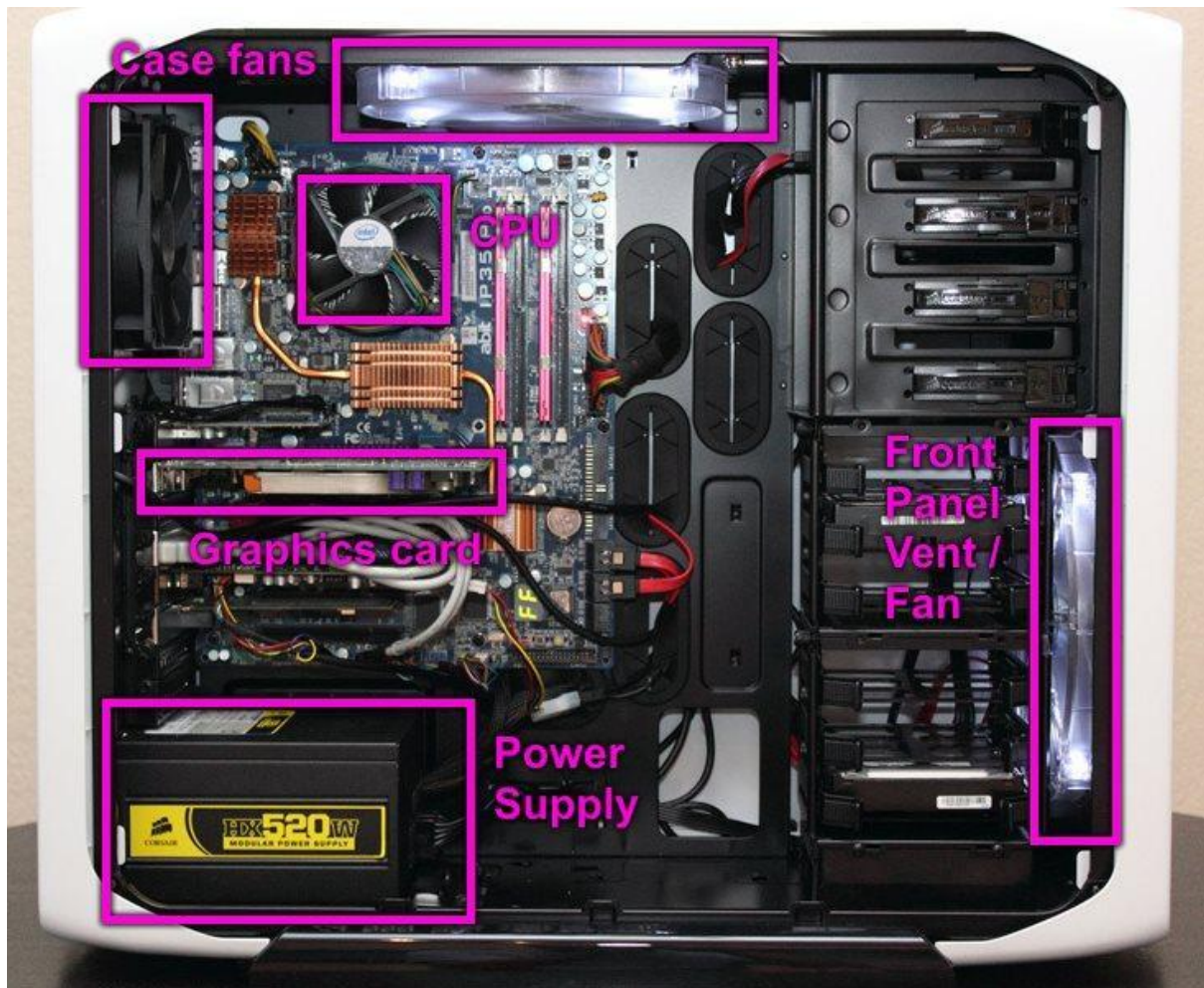
1. Shut down the computer.
2. Disconnect the power cable.
3. Open the side panel carefully.
4. Locate the fans, heatsinks and GPU cooling system.

If you're using a laptop, don't open it unless you're authorized to do so. You can instead use a hardware simulator or a clear internal photo.

Step 2: Take a clear photo 📷

Take a photo showing the cooling components clearly.





Step 3: Label the main components

You can label these components on your photo:

Component	Function
CPU Fan	Spins to move air across the CPU heatsink and remove heat.
CPU Heatsink	Absorbs heat from the CPU and transfers it to the surrounding air.
GPU Fans	Move air through the GPU's cooling system to remove heat.

Component	Function
GPU Heatsink	Absorbs and spreads heat generated by the GPU.
Case Fans	Move cool air into the case and hot air out of it.
Heat Pipes	Transfer heat from the GPU/CPU area to the heatsink fins.
Radiator	In a liquid-cooling system, releases heat from the coolant into the air.
Pump	Circulates coolant through a liquid-cooling system.

3.

Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Factor	Air Cooling	Liquid Cooling
Cost	Generally cheaper	Generally more expensive
Cooling efficiency	Good for most GPUs and normal gaming	Can provide very strong heat transfer and lower temperatures in suitable setups
Maintenance	Simple; mainly requires dust cleaning	More complex; includes pump, tubes and radiator that need attention

Factor	Air Cooling	Liquid Cooling
Installation	Easier to install	More complicated to install
Noise	Fans can become noisy at high speeds	Can be quieter in some setups, but radiator fans and pump still make noise
Components	Mainly heatsink and fans	Cooling block, pump, coolant, tubes, radiator and fans
Risk	Fewer components, so fewer possible failure points	More components; pump failure or coolant issues are additional concerns
Best suited for	Most gaming PCs and general-purpose systems	High-performance or enthusiast PCs where extra cooling capacity is useful

4.

Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

Choice: Air Cooling

I would choose air cooling for the GPU because it is generally cheaper, simpler, and easier to maintain than liquid cooling. A good-quality air cooler can handle the high temperatures produced during graphics-heavy gaming while keeping the GPU stable. Streaming IPL matches does not require extreme GPU cooling by itself, so liquid cooling would be unnecessary for this

build. Therefore, air cooling gives a good balance of performance, cost, reliability, and maintenance.